CAUTION: High voltages are present in this laboratory exercise! Do not make or modify any connections with the power on unless otherwise specified! If any danger occurs, immediately press the RED EMERGENCY OFF button on the Three-phase Power Supply Module. Complete this laboratory exercise as quickly as possible to avoid the rise in temperature under load condition. The machines are heavy, make sure to practice safety precautions when moving them from the cabinet to the table top.

Lab 2: Three-phase Rotor Winding Induction Motor

Activity 2.1: Blocked-Rotor Test

2.1.1. Aim of activity

After completing this exercise, you should be able to demonstrate the blocked-rotor characteristic of a three-phase rotor winding induction motor.

2.1.2. Equipment required

Table 2.1.1. List of equipment

Qty	Description	Cat. No.
1	Three-phase Rotor Winding Motor	EM-3330-3B
1	Magnetic Powder Brake Unit	EM-3320-1C
1	Brake Controller	EM-3320-1N
1	System Transformer	EM-3340-3B
1	Three-phase Power Supply Module	EM-3310-1E
1	3ϕ AC/DC Power Supply	EM-3310-1D
1	Three-pole Current Limit Protection Switch Module	EM-3310-2A
1	Four-pole Switch Module	EM-3310-2B
1	Digital Power Analysis Meter	EM-3310-3H
1	Fuse Set	EM-3310-5B
1	Coupling	EM-3390-2A
1	Connecting Leads Set	EM-3390-3A

1	Safety Bridging Plugs Set	EM-3390-4A

2.1.3. Schematic

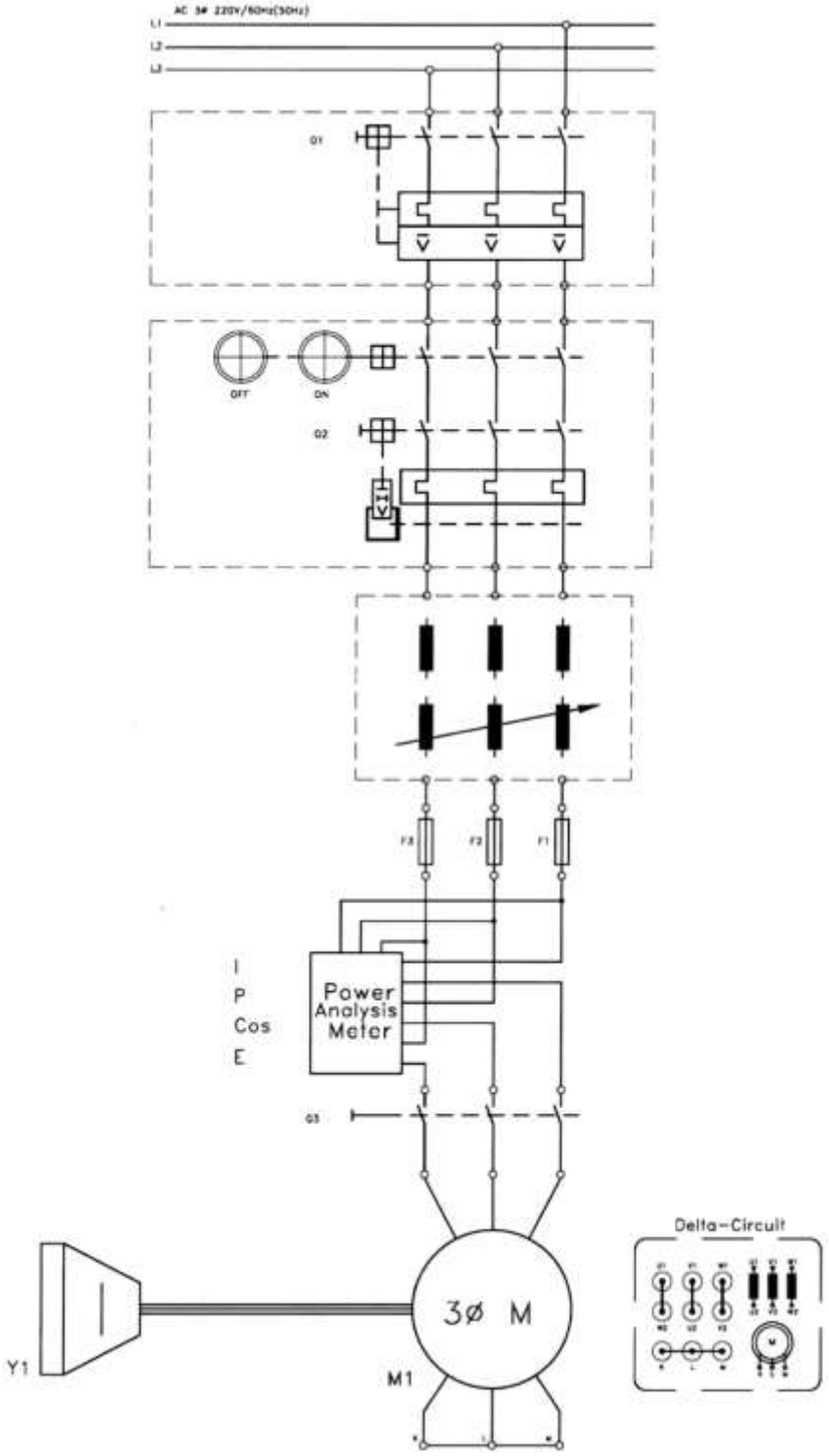


Figure 3.1.1. Circuit diagram for blocked-rotor test

2.1.4. Connection

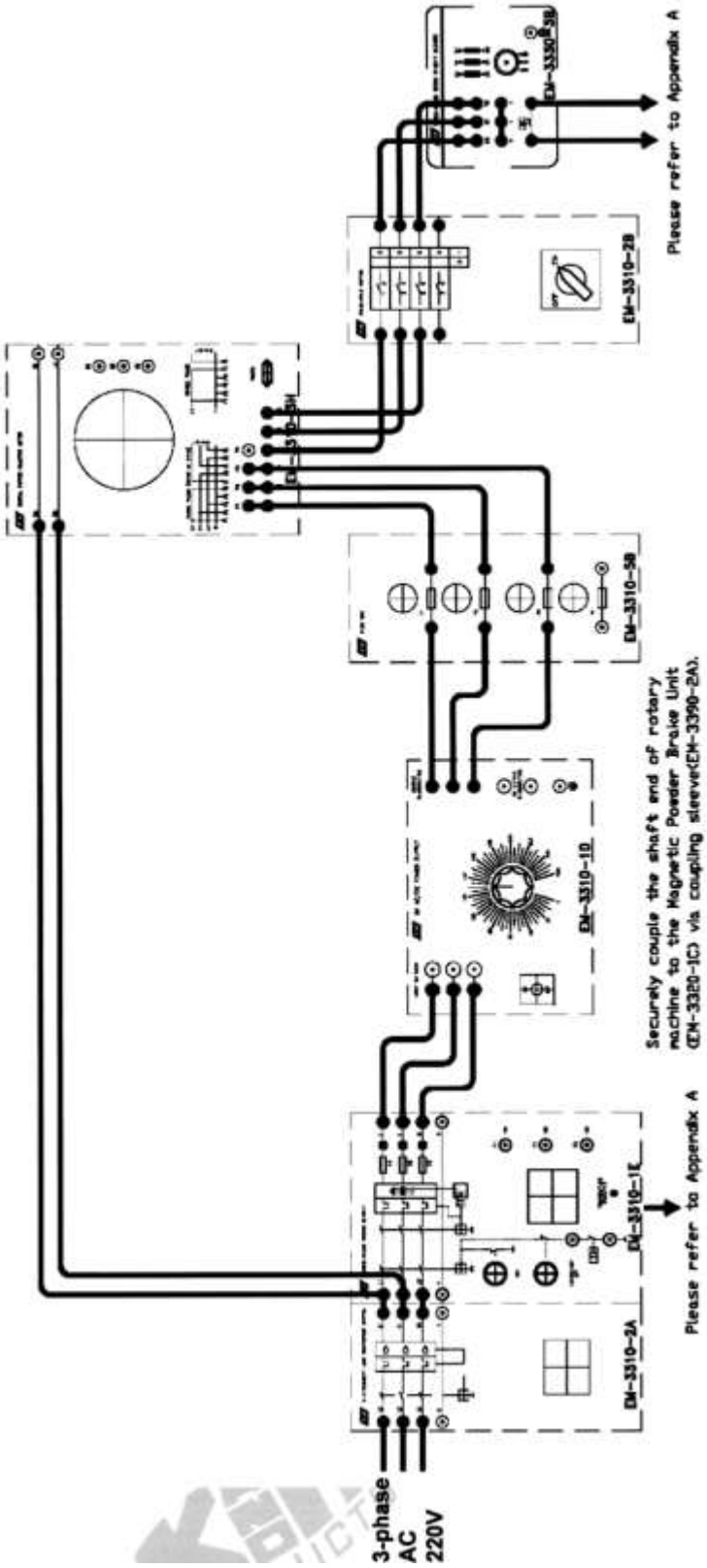


Figure 2.1.2. Connection diagram for blocked-rotor test

2.1.5. Experimental procedure

1. Place the Three-phase Rotor Winding Motor, Magnetic Powder Brake Unit, Brake Controller and 3ϕ AC/DC Power Supply on the Laboratory Table. Mechanically couple the Three-phase Rotor Winding Motor to the Magnetic Powder Brake Unit using a Coupling. Securely lock Machine Bases together using delta screws. Electrically connect the Brake Controller to the Magnetic Powder Brake Unit using the supplied cable.

2. Install the required Modules in the Experimental Frame. Construct the circuit in accordance with the circuit diagram and the connection diagram in Figure 2.1.1 and 2.1.2, respectively. Have the instructor or lab technician to check on your completed circuit. **Note: The thermal switches of Three-phase Power Supply Module, Three-Phase Wound Rotor Induction Motor and Magnetic Powder Brake Unit must be connected together.**

3. Set the on-off switch on Four-pole Switch Module to the OFF position. Set the control knob voltage on the 3ϕ AC/DC Power Supply to the 0 position. Sequentially turn on the main switches, System Transformer, Brake Controller, Magnetic Powder 3rake Unit, 3-P Current Limit Protection Switch, Three-phase Power Supply, and 3ϕ AC/DC Power Supply Modules.

4. Change the 7-segment LED display on the Brake Controller to display Power and Voltage values: LED\P&V. Then manipulate the Brake Controller to operate in Mode\Open Loop\Manual mode and set the output brake voltage to 3.5 V.

5. Set the on-off switch on Four-pole Switch Module to ON position for starting the motor in delta, the motor should not run.

6. On the 3ϕ AC/DC Power Supply, slowly turn the voltage control knob clockwise so that the motor current I is equal to the rated value of 2.0 A.

7. Record the motor current I, motor voltage E, motor power P, and power factor cos θ values displayed by the Digital Power Analysis Meter as quickly as possible in Table 2.1.2 in the Results and Discussions section.

8. Sequentially turn off the Four-pole Switch, 3ϕ AC/DC Power Supply, Three-phase Power Supply, 3-P Current Limit Protection Switch Modules, System Transformer, Magnetic Powder Brake Unit, and Brake Controller.

2.1.6. Results and discussions

1. Fill in the values obtained from the previous section to Table 2.1.2 below:

Table 2.1.2. Measured values of I, E, P and cos θ

I (A)	P (W)	E (V)	cos θ
2.0			

2. Discuss and conclude your findings from this activity.

Activity 2.2: Torque-Speed Characteristic

2.2.1. Aim of activity

After completing this exercise, you should be able to demonstrate the torque-speed characteristic of a three-phase rotor winding induction motor.

2.2.2. Equipment required

Table 2.2.1. List of equipment

Qty	Description	Cat. No.
1	Three-phase Rotor Winding Induction Motor	EM-3330-3B
1	Magnetic Powder Brake Unit	EM-3320-1C
1	Brake Controller	EM-3320-1N
1	System Transformer	EM-3340-3B
1	Three-phase Power Supply Module	EM-3310-1E
1	Three-pole Current Limit Protection Switch Module	EM-3310-2A
1	Digital Power Analysis Meter	EM-3310-3H
1	Fuse Set	EM-3310-5B
1	Coupling	EM-3390-2A
1	Connecting Leads Set	EM-3390-3A
1	Safety Bridging Plugs Set	EM-3390-4A

2.2.3. Schematic

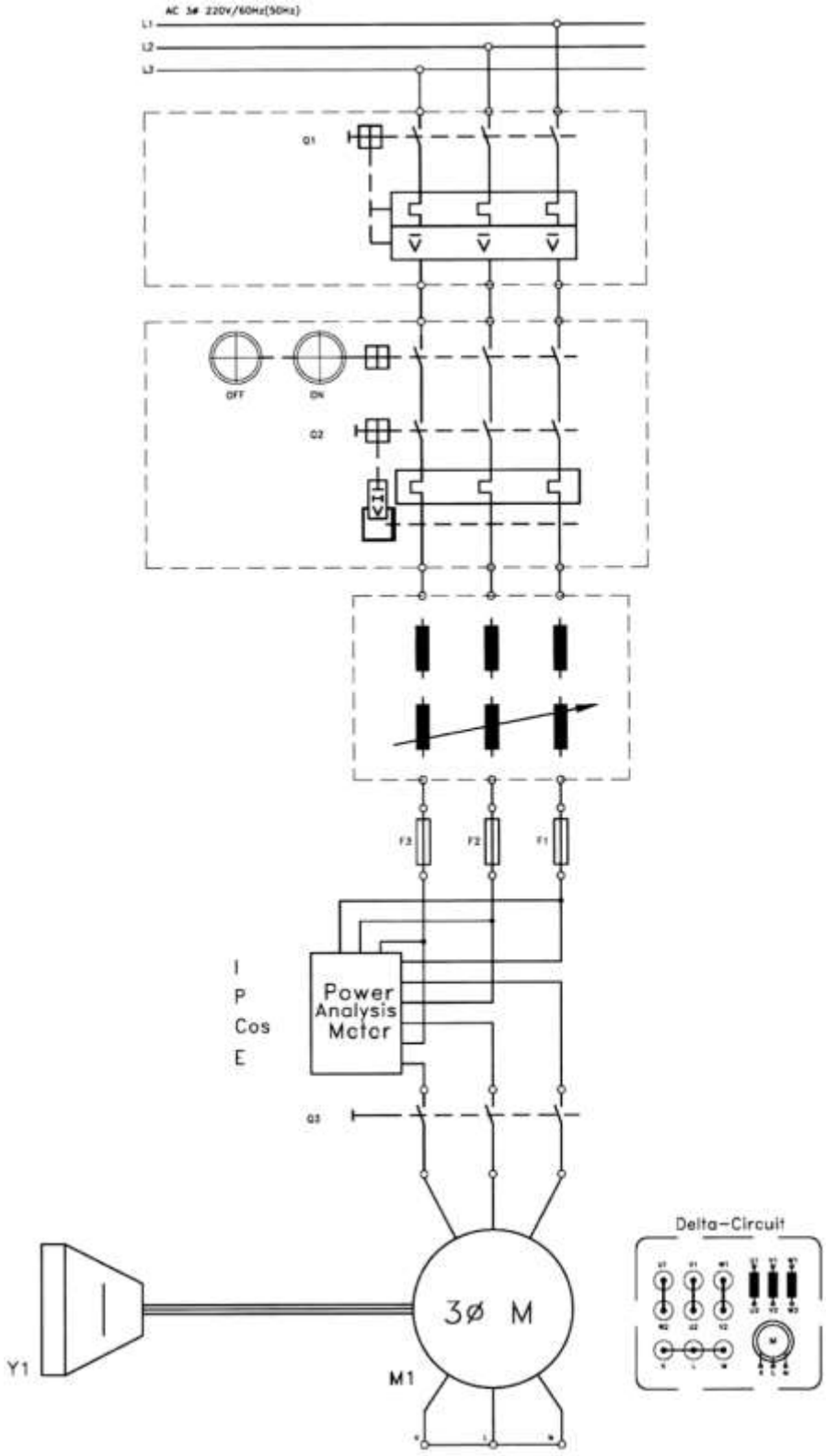


Figure 2.2.1. Circuit diagram for torque-speed characteristic test

2.2.4. Connection

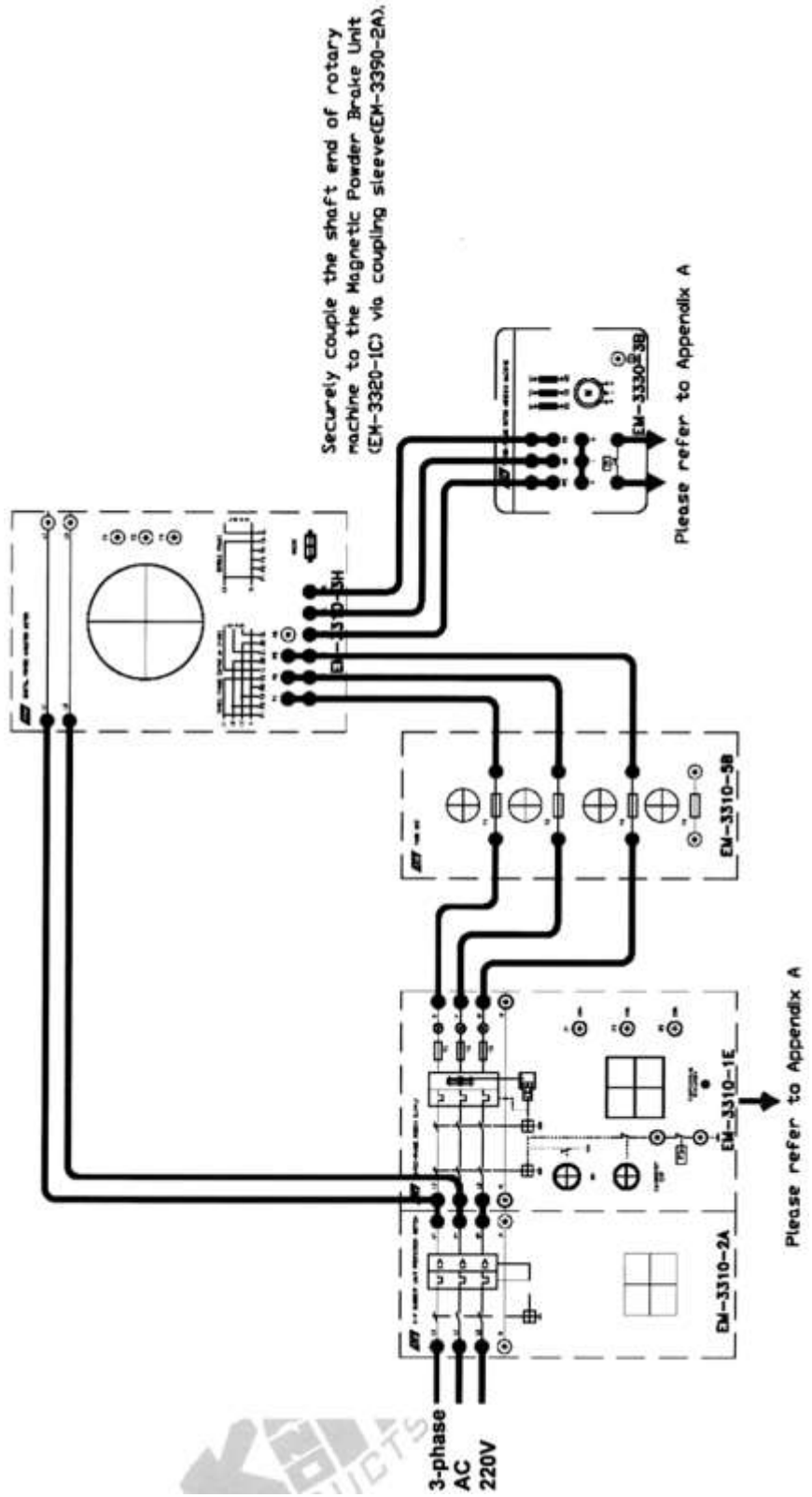


Figure 2.2.2. Connection diagram for torque-speed characteristic test

2.2.5. Experimental procedure

1. Place the Three-phase Rotor Winding Motor, Magnetic Powder Brake Unit, and Brake Controller on the Laboratory Table. Mechanically couple the Three-phase Rotor Winding Motor to the Magnetic Powder Brake Unit using a Coupling. Securely lock Machine Bases together using delta screws. Electrically connect the Brake Controller to the Magnetic Powder Brake Unit using the supplied cable.

2. Install the required Modules in the Experimental Frame. Construct the circuit in accordance with the circuit diagram and the connection diagram in Figure 2.2.1 and 2.2.2, respectively. Have the instructor or lab instructor to check on your completed circuit**. Note: The thermal switches of Three-phase Power Supply Module, Three-Phase Rotor Winding Induction Motor and Magnetic Powder Brake Unit must be connected together**.

3. Sequentially turn on the main switches, System Transformer, Brake Controller, Magnetic Powder Brake Unit, Three-phase Power Supply and 3-P Current Limit Protection Switch Modules. The motor should start running in delta.

4. Manipulate the Brake Controller to operate in Model\Closed Loop\Constant Torque mode and set the output torque to 0 kg-m. If the Controller doesn't operate normally, reboot it by pressing the RESET button. If the rotor is locked by a heavy brake torque, release the braking by pressing the BACK button.

5. Record the motor power P, motor current l, motor speed and power factor cos θ values displayed by the Digital Power Analysis Meter in Table 2.2.2 in the Results and Discussions section.

6. Manipulate the Brake Controller to release the braking. That is to say, release the braking by pressing the ESC or BACK button on the Brake Controller.

7. Repeat steps 4 through 6 for other torque settings listed in Table 2.2.2. **Note: The motor current must not exceed 130% of the rated current (*2 A* × *1.3* = *2.6 A*).**

8. Sequentially turn off the Three-phase Power Supply, 3-P Current Limit Protection Switch Modules, System Transformer, Magnetic Powder Brake Unit and Brake Controller.

2.2.6 Results and discussions

1. Fill in the values obtained from the previous section to Table 2.2.2 below:

Table 2.2.2. Measured I, E, P, and cos θ values

T (kg-m)	0	0.05	0.1	0.15	0.2	0.25	0.3
P (W)							
I (A)							
cos θ							
N (rpm)							

2. Using the results of Table 2.2.2, plot the T vs. N, T vs. s, cos θ vs. P, and N vs. P curves. Analyse the relationship between these parameters with relevant theories and equations.

3. State the conclusion of this activity.

Activity 2.3: Starting Characteristic

2.3.1. Aim of activity

After completing this exercise, you should be able to demonstrate the starting characteristic of a three-phase rotor winding induction motor.

2.3.2. Equipment required

Table 2.3.1. List of equipment

Qty	Description	Cat. No.
1	Three-phase Rotor Winding Induction Motor	EM-3330-3B
1	Magnetic Powder Brake Unit	EM-3320-1C
1	Brake Controller	EM-3320-1N
1	System Transformer	EM-3340-3B
1	Three-phase Power Supply Module	EM-3310-1E
1	Three-pole Current Limit Protection Switch Module	EM-3310-2A
1	Four-Pole Switch	EM-3310-2B
1	Digital Power Analysis Meter	EM-3310-3H
1	Winding Machine Starter	EM-3310-4E
1	Coupling	EM-3390-2A
1	Connecting Leads Set	EM-3390-3A
1	Safety Bridging Plugs Set	EM-3390-4A

2.3.3. Schematic

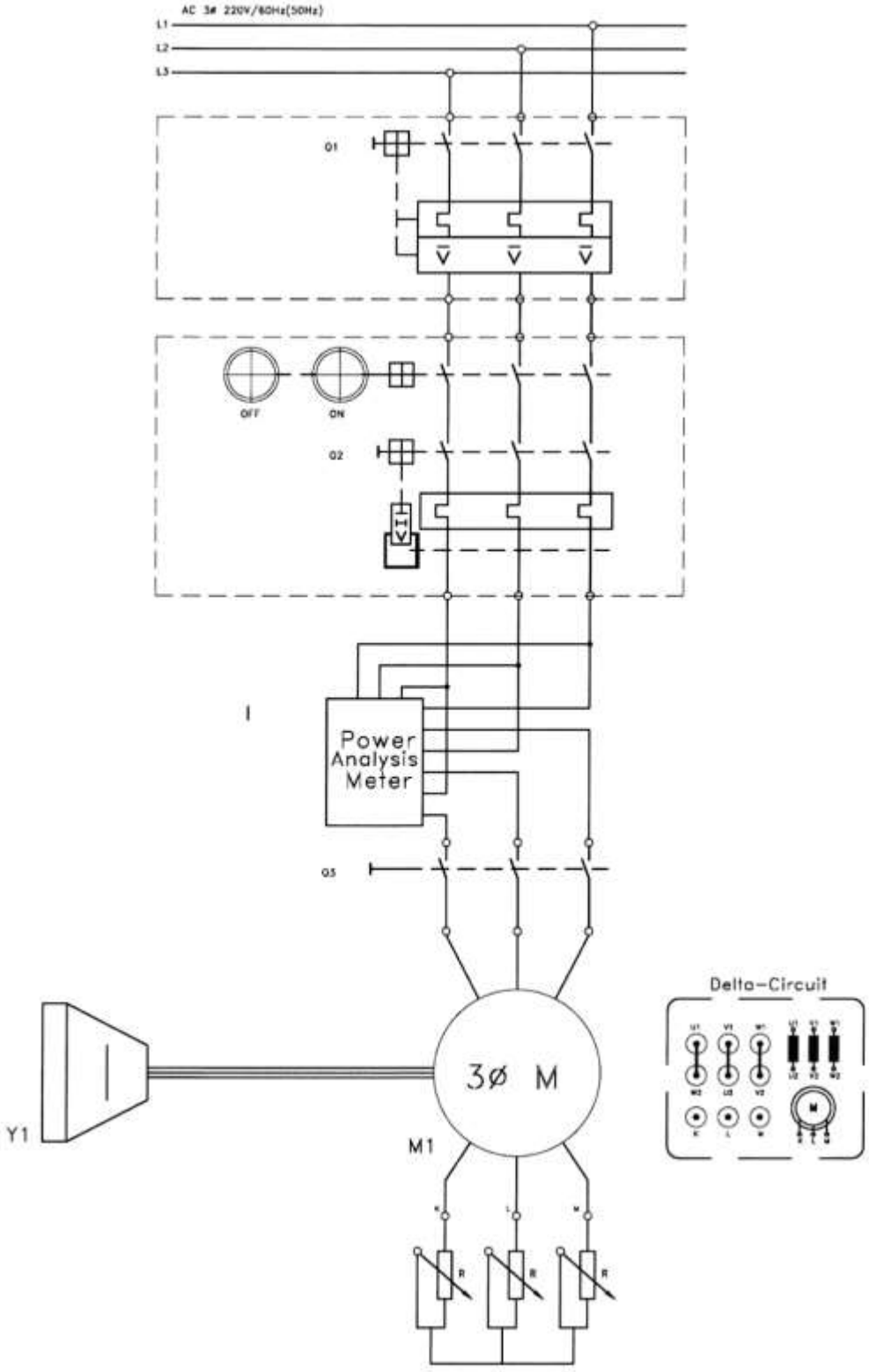


Figure 2.3.1. Circuit diagram for starting characteristic test

2.3.4. Connection

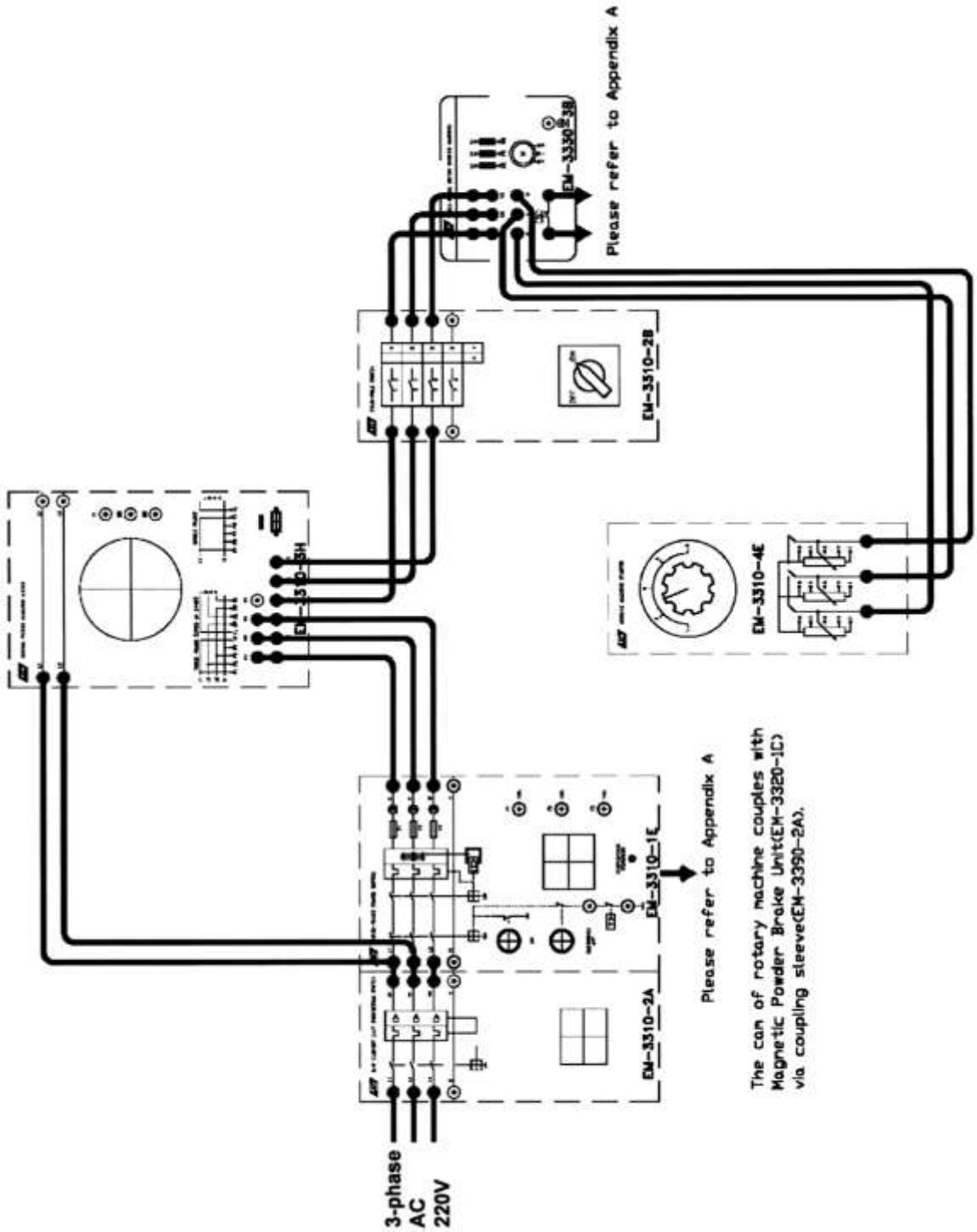


Figure 2.3.2. Connection diagram for starting characteristic test

2.3.5. Experimental procedure

1. Place the Three-phase Rotor Winding Motor, Magnetic Powder Brake Unit, and Brake Controller on the Laboratory Table. Mechanically couple the Three-phase Rotor Winding Motor to the Magnetic Powder Brake Unit using a Coupling. Securely lock Machine Bases together using delta screws. Electrically connect the Brake Controller to the Magnetic Powder Brake Unit using the supplied cable.

2. Install the required Modules in the Experimental Frame. Construct the circuit in accordance with the circuit diagram and the connection diagram in Figure 2.3.1 and 2.3.2, respectively. Have the instructor or lab technician to check on your completed circuit. **Note: The thermal switches of Three-phase Power Supply Module, Three-Phase Rotor Winding Induction Motor and Magnetic Power Brake unit must be connected together.**

3. Change the 7-segment LED display on the Brake Controller to display Power and Voltage values: LED\P&V. Then manipulate the Brake Controller to operate in Mode\Open Loop\Manual mode and set the output brake voltage to 10 V. Place the knob on the Winding Machine Starter in 1 position.

4. Sequentially tum on the main switches, System Transformer, Brake Controller, Magnetic Powder Brake Unit, 3-P Current Limit Protection Switch, Three-phase Power Supply, and Four-Pole Switch Modules to start the Three-phase Rotor Winding Motor. The motor should be at standstill. Record the highest motor current I displayed by the Digital Power Analysis Meter in Table 2.3.2 in the Results and Discussions section. **Note: The motor current is large so complete this exercise as quickly as possible.**

5. Turn off the Four-Pole Switch and Three-phase Power Supply, then repeat the experiment with different knob positions listed in Table 2.3.2.

6. Before starting the next experiment, sequentially turn off the Four-Pole Switch, Three-phase Power Supply, 3-P Current Limit Protection Switch Modules, System Transformer, Magnetic Powder Brake Unit and Brake Controller.

7. Manipulate the Brake Controller to operate in Mode\Closed Loop\Constant Torque mode and set the output torque to 0.3 kq-m. If the Controller doesn't operate normally, reboot it by pressing the RESET button. If the rotor is locked by a heavy brake torque, release the braking by pressing the BACK button.

8. Place the knob on the Winding Machine Starter in 1 position.

9. Sequentially turn on the main switches, System Transformer, Brake Controller, Magnetic Powder Brake Unit, 3-P Current Limit Protection Switch, Three-phase Power Supply, and Four-Pole Switch Modules to start the Three-phase Rotor Winding Motor. The motor should run at low speed.

10. Record the motor current I and motor speed N values displayed by the Digital Power Analysis Meter in Table 2.3.3 in the Results and Discussions section. **Note: Complete this part as quickly as possible to avoid the rise in temperature under load condition.**

11. Sequentially turn off the Four-Pole Switch, Three-phase Power Supply and 3-P Current Limit Protection Switch Modules, System Transformer, Magnetic Powder Brake Unit, and Brake Controller.

12. Repeat Steps 8 through 10 for other knob positions listed in Table 2.3.3.

2.3.6 Results and Discussions

1. Fill in the values obtained from the previous section to Table 2.3.2 and 2.3.3 below:

Table 2.3.2. Measured motor current I

Knob position	1	2	3	4	5
I (A)					

Table 2.3.3. Measured motor current I and motor speed N

Knob position	1	2	3	4	5
I (A)					
N (rpm)					

2. Analyse the result and elucidate the relationship between the parameters.

3. Using the results of Table 2.3.3, sketch and compare the T vs. s curve of each resistance and show that the motor speed can be controlled via the adjustment of external resistance. (Tips: sketch the curves based on how the resistance will affect the curve, the speed/slip has to follow the table, the rest is based on estimation)

4. Is the resistance of the Winding Machine Starter highest or lowest at knob position 1? Explain.

5. State the conclusion of this activity.

Note: Consolidate all the lab activities 2.1, 2.2, and 2.3 into a single report.

APPENDIX 1

MARKING RUBRICS

Component Title	Lab report				Percentage (%)	100	
Criteria	Score and Descriptors					Weight (%)	Marks
	Excellent (5)	Good (4)	Average (3)	Need Improvement (2)	Poor (1)		
Formatting, quality of figures and plots	All sections are presented in a neat and tidy manner with consistent format and style. Complete captions for tables and figures are given and mentioned in the main text. All figures are clear with important labels and comments. Title and the axis of all plots are well labelled, clear, and easy to read.	Most sections are presented in a neat and tidy manner with consistent format and style. Complete captions for tables and figures are given and most are mentioned in the main text. Most figures are clear with labels and comments. Title and the axis of most plots are well labelled and clear.	Some sections are inconsistent in terms of format and style. Most captions for tables and figures are given and mentioned in the main text. Some figures and components in the circuit diagram are not clear and missing important labels and comments. Some title and axis of plots are not well labelled and clear.	Most sections are inconsistent in terms of format and style. Incomplete captions for tables and figures and most are not mentioned in the main text. Most figures and components in the circuit diagram are not clear and missing important labels and comments. Most title and axis of plots are not well labelled and clear.	Messy report and difficult to read with inconsistent format and style. No captions for tables and figures provided. Blurry figures and plots with missing labels provided. Figures and plots are difficult to read and understand.	15	
Introduction and Methodology	Excellently written, concise introduction and methodology with important information that are relevant to the topic.	Well written, concise introduction and methodology with important information that are relevant to the topic.	Averagely written, concise introduction and methodology, consists some irrelevant information.	Averagely written, lengthy introduction and methodology, consists many irrelevant information.	Poorly written, lengthy introduction and methodology, consists many irrelevant information.	15	
Results and discussions	Completed all experiments and data collection. All measured data, calculations, and graphs/plots are correct. Able to evaluate the results with prudent judgment. Analysis is done with excellent description of the relationship involved and explanation with relevant theories.	Completed all experiments and data collection. Most measured data, calculations, and graphs/plots are correct. Results are compared and analysed with good description of the relationship involved and with relevant theories.	Completed most experiments and data collection. Some measured data, calculations, and graphs/plots are incorrect. Results analysis are done with adequate description of the relationship involved.	Completed most experiments, but most measured data, calculations, and graphs/plots are incorrect. Only restatement of the results without commenting on the expected key points.	Experiments not completed. Most measured data, calculations, and graphs/plots are incorrect. Discussion shows little understanding of what the experiment is all about.	50	
Conclusion	The aim of the experiment has been achieved, the key features of the methods used and most important results are summarized.	The aim of the experiment has been achieved, most key features of the methods used and results are summarized.	A conclusion is drawn, but only supported by some experimental evidence.	A conclusion is drawn, but not supported by any experimental evidence.	No sensible conclusion. No comparisons and evidence.	20	
					TOTAL	**100**	

Note to students: Please attach this appendix together with the submission of coursework.